Milk Hill 2001: Actual Image-Derived Mathematics and Binary Test

This report records the calculation actually performed on the clean OH 35mm photograph contained in the supplied Temporary Temples PDF. It does not treat the earlier English phrases as proven. Every numerical output below comes from the supplied image and the fixed rules stated here.

Important limitation: the PDF contains a web-page thumbnail, not the original 35 mm scan. At this resolution the automated segmentation recovered 361 measurable bright components, not all 409 reported circles. Therefore this is a complete calculation on the available image, but not a final 409-circle forensic survey.

Source photograph used for measurement: Milk Hill, Wiltshire, 12 August 2001, Wheat OH 35mm.

1. Image segmentation

The photograph was converted from RGB to HSV. A pixel was classified as part of a bright flattened circle when its HSV value channel exceeded 199:

```
M(x,y) = 1 if V(x,y) > 199; otherwise M(x,y) = 0
```

Eight-connected components were then measured. Components were retained when 3 <= area <= 3000 pixels and both bounding-box dimensions were below 100 pixels.

Recovered measurable components: **361**

Equivalent diameter for each component:

```
d_i = 2 * sqrt(A_i / pi)
```

Red outlines show the measured connected components. Every fifth sequence number is displayed to reduce clutter.

2. Coordinate normalization

Because the photograph is oblique, raw image coordinates were centered and whitened using the covariance eigenvectors. This reduces the dominant elliptical perspective distortion without claiming a full photogrammetric reconstruction.

```
p_i = [x_i, y_i]^T

q_i = Lambda^(-1/2) Q^T (p_i - mean(p))

r_i = sqrt(q_x,i^2 + q_y,i^2)

theta_i = atan2(q_y,i, q_x,i)
```

Image centroid used: (478.326, 298.342) pixels. Covariance eigenvalues: 28740.616, 16192.120.

3. Six-sector assignment

A phase offset was searched over one 60-degree interval. For each phase, points were assigned to one of six equal angular sectors. The selected phase minimized the variance of the six component counts.

```
a_i = floor( ((theta_i - phi) mod 2*pi) / (pi/3) )
```

Selected phase phi = 0.993458 radians = 56.921 degrees. Sector counts = [61, 60, 60, 57, 58, 65]. Count variance = 6.472222.

4. Fixed binary conversion rule

Within each sector, components were sorted from the normalized center outward. The binary rule was fixed before interpreting the output:

```
b_i = 1 if d_i > d_(i-1); otherwise b_i = 0
```

The first component in each sector has no predecessor and contributes no bit. With 361 detected components across six sectors, this produced 355 bits.

First 120 bits:

```
10010011001010001001110101001001011111000011010010100100110110000011010101101010111011001001
1100101101100110100100000001
```

5. Five-bit grouping and letter test

Bits were divided into non-overlapping five-bit groups. Each group was converted to decimal:

```
v = 16b1 + 8b2 + 4b3 + 2b4 + b5
```

For a transparent test only, values 1-26 were mapped to A-Z, 0 to SPACE, and 27-31 were retained as control values. No values were altered to force words.

Group	Bits	Decimal	Interpretation
1	10010	18	R
2	01100	12	L
3	10100	20	T
4	01001	9	I
5	11010	26	Z
6	10010	18	R
7	01011	11	K
8	11100	28	CONTROL_28
9	00110	6	F
10	10010	18	R

11	10010	18	R
12	01101	13	M
13	10000	16	P
14	01101	13	M
15	01011	11	K
16	01010	10	J
17	11101	29	CONTROL_29
18	10010	18	R
19	01110	14	N
20	01011	11	K
21	01100	12	L
22	11010	26	Z
23	01000	8	H
24	00001	1	A
25	01010	10	J
26	11001	25	Y
27	01101	13	M
28	01010	10	J
29	11010	26	Z
30	10101	21	U

Complete direct output:

```
RLTIZRK[28]FRRMPMKJ[29]RNKLZHAJYMJZUVLKFSFUFCJZXKYEJTTR[28]TJRIVUJ[30]MKGNYJZRZYX V
```

Result: this fixed, reproducible diameter-comparison encoding does not produce the earlier claimed sentence. It produces the sequence shown above. That finding is important because it distinguishes measured output from a desired interpretation.

6. First 40 measured rows

sequence_id	arm	rank_center_outward	x_px	y_px	area_px	equivalent_diameter_px	radius_normalized	theta_degrees	binary_bit
1	0	1	475.750	269.250	4	2.257	0.229	85.641	
2	0	2	479.200	266.600	5	2.523	0.250	90.630	1
3	0	3	466.500	267.000	4	2.257	0.255	73.600	0
4	0	4	483.667	264.333	3	1.954	0.269	96.163	0
5	0	5	469.641	248.957	117	12.205	0.391	81.918	1
6	0	6	506.333	250.000	9	3.385	0.415	112.882	0
7	0	7	484.333	230.667	3	1.954	0.533	93.256	0
8	0	8	490.500	231.000	6	2.764	0.535	97.166	1
9	0	9	465.027	229.235	260	18.195	0.548	81.217	1
10	0	10	446.750	229.250	4	2.257	0.573	70.466	0
11	0	11	524.000	224.000	3	1.954	0.645	114.129	0
12	0	12	465.980	204.332	346	20.989	0.742	83.812	1
13	0	13	488.250	200.750	4	2.257	0.770	93.808	0
14	0	14	445.100	198.100	10	3.568	0.810	75.449	1
15	0	15	440.100	197.400	10	3.568	0.823	73.545	0
16	0	16	501.000	184.500	6	2.764	0.906	97.940	0
17	0	17	506.000	185.000	5	2.523	0.907	99.818	0
18	0	18	477.064	179.152	486	24.876	0.937	88.990	1
19	0	19	390.667	189.500	6	2.764	0.996	58.175	0
20	0	20	466.500	169.500	4	2.257	1.014	85.502	0
21	0	21	523.200	164.500	10	3.568	1.087	103.547	1
22	0	22	533.900	165.450	20	5.046	1.097	106.835	1
23	0	23	501.737	151.622	529	25.953	1.162	96.270	1
24	0	24	388.600	164.200	5	2.523	1.176	62.698	0
25	0	25	379.667	160.667	24	5.528	1.225	61.072	1
26	0	26	555.250	152.583	12	3.909	1.235	110.999	0
27	0	27	371.185	160.778	27	5.863	1.248	59.024	1
28	0	28	564.118	153.294	17	4.652	1.251	113.316	0
29	0	29	406.200	144.600	5	2.523	1.278	69.998	0
30	0	30	489.297	130.108	37	6.864	1.324	92.247	1
31	0	31	397.636	140.091	11	3.742	1.328	68.447	0
32	0	32	388.333	138.111	9	3.385	1.363	66.521	0
33	0	33	538.567	130.272	533	26.051	1.370	104.476	1
34	0	34	432.667	123.333	3	1.954	1.399	78.349	0
35	0	35	492.500	120.000	4	2.257	1.405	92.858	1
36	0	36	457.545	111.091	11	3.742	1.476	84.681	1

sequence_id	arm	rank_center_outward	x_px	y_px	area_px	equivalent_diameter_px	radius_normalized	theta_degrees	binary_bit
37	0	37	426.611	114.111	18	4.787	1.477	77.529	1
38	0	38	419.185	113.926	27	5.863	1.488	75.887	1
39	0	39	356.019	128.558	1068	36.876	1.512	60.946	1
40	0	40	594.401	120.359	853	32.956	1.562	115.447	0

The accompanying CSV contains all 361 component rows. The separate binary-groups CSV contains all five-bit groups and decimal conversions.

7. What would be required for the complete 409-circle result

The original full-resolution OH 35mm scan is required. With that scan, the same pipeline can be rerun, components manually verified, and the six missing first-of-sector bits handled under an explicitly documented framing rule. The present files preserve the full calculation performed on the supplied evidence and do not fabricate missing measurements.